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(54) ELECTRONIC CONTROL UNIT

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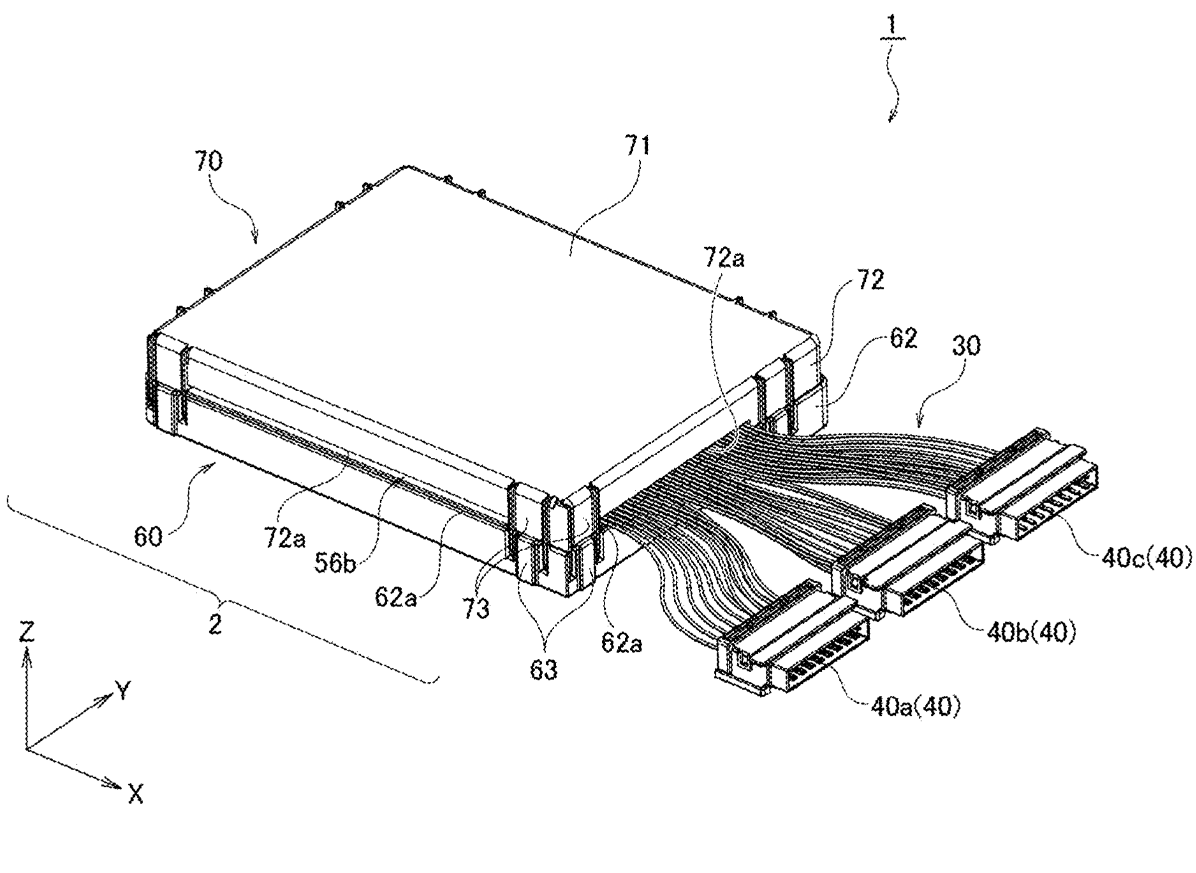
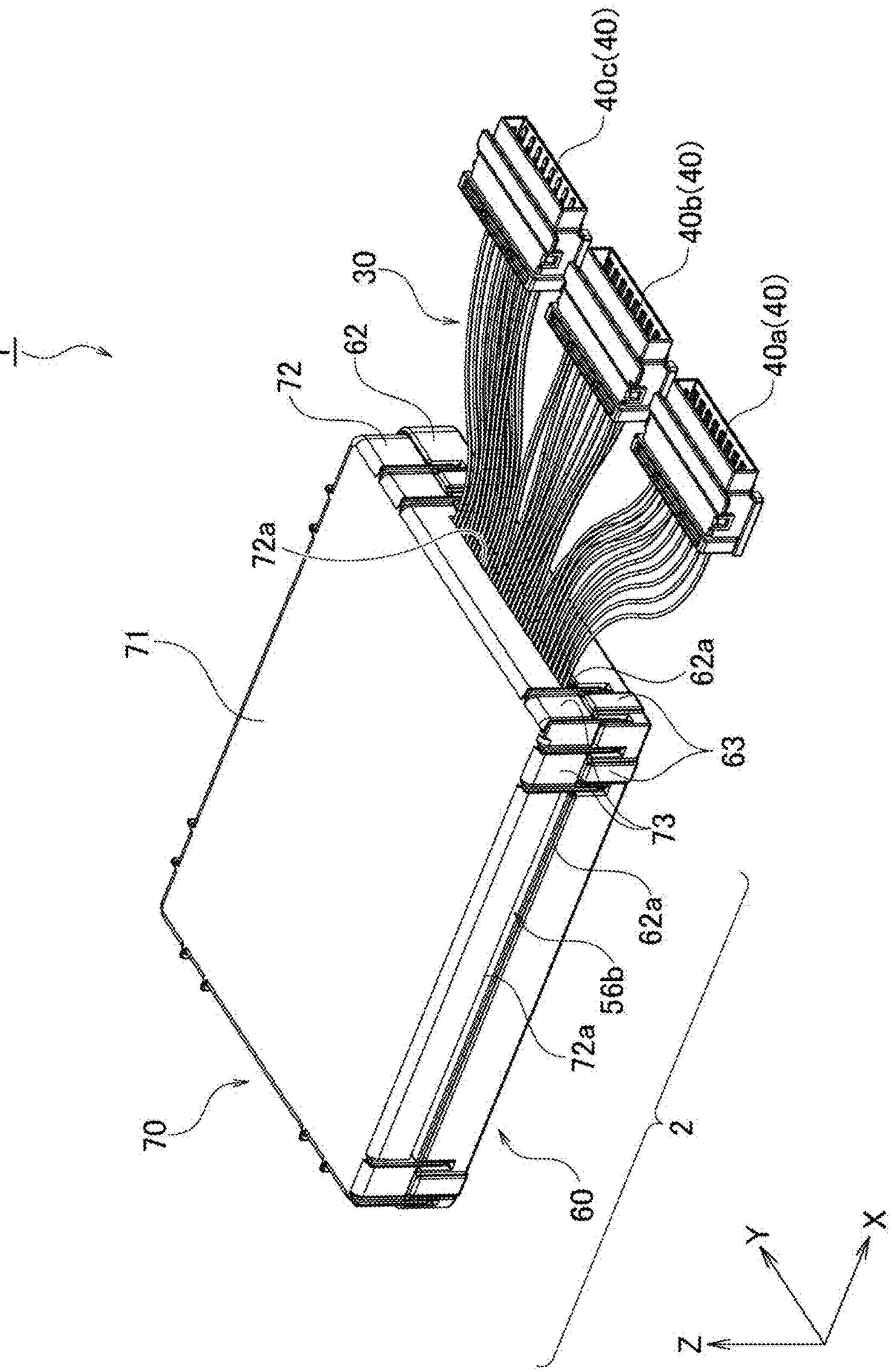
(57) **ABSTRACT**

In an electronic control unit, first terminal blocks on a routing board have an opening groove, which is penetrating in a first direction orthogonal to a normal direction of a routing surface, and inside of which a part of a second end of terminals is exposed. At least a part of an upper surface of the first terminal block is a routing region surface. A height position of the routing region surface from the routing surface is set higher than a bottom surface of the opening groove, and lower than a height position of a tip edge of the side wall by a distance that is larger than a dimension value of an outer diameter of the electric wires. A third width of the routing region surface in the first direction is set larger than the dimension value of the outer diameter of the electric wires.

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30
40c(40)
40b(40)
40a(40)
62
72
72a
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62a
63
73
62a
56b
72a
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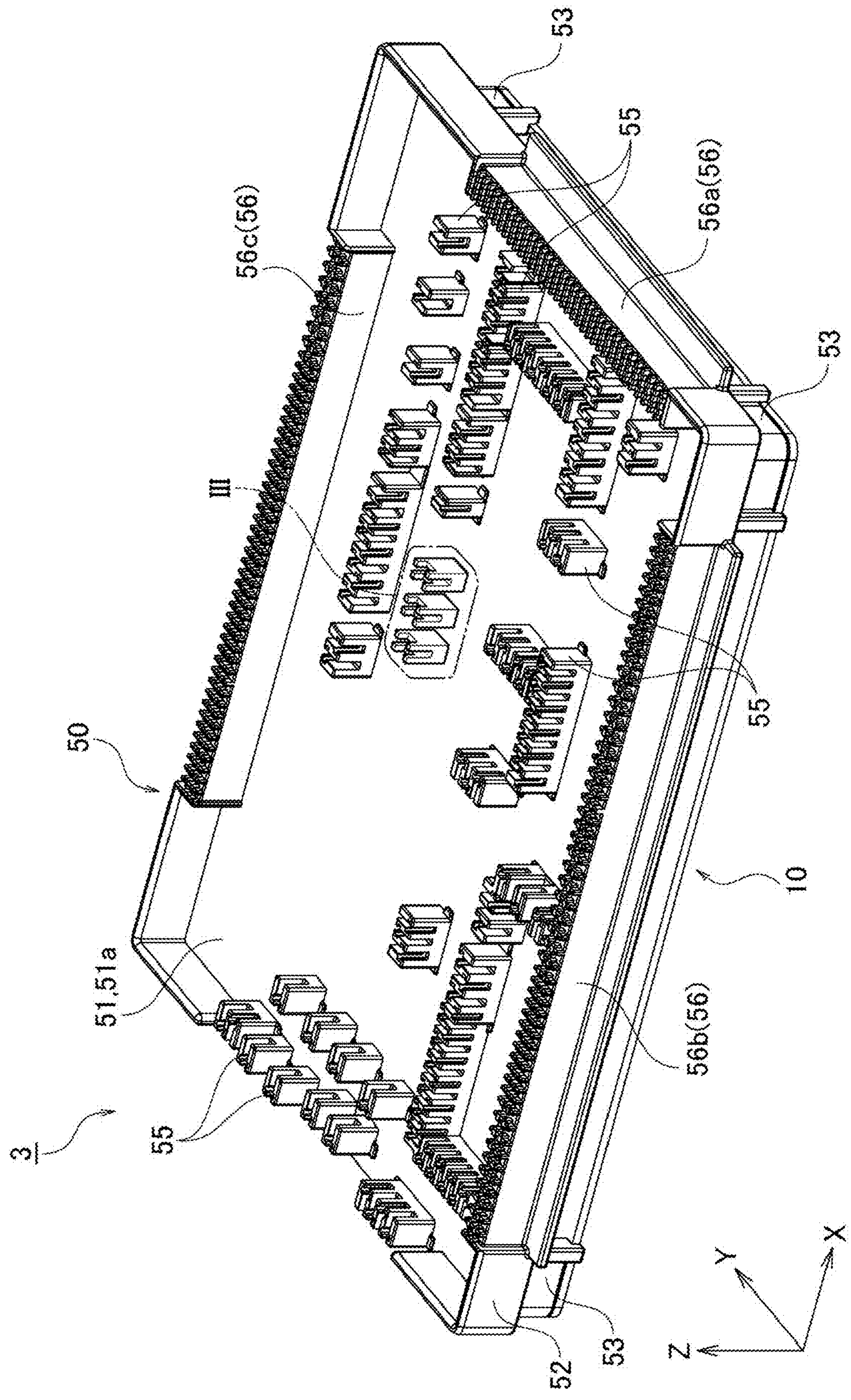
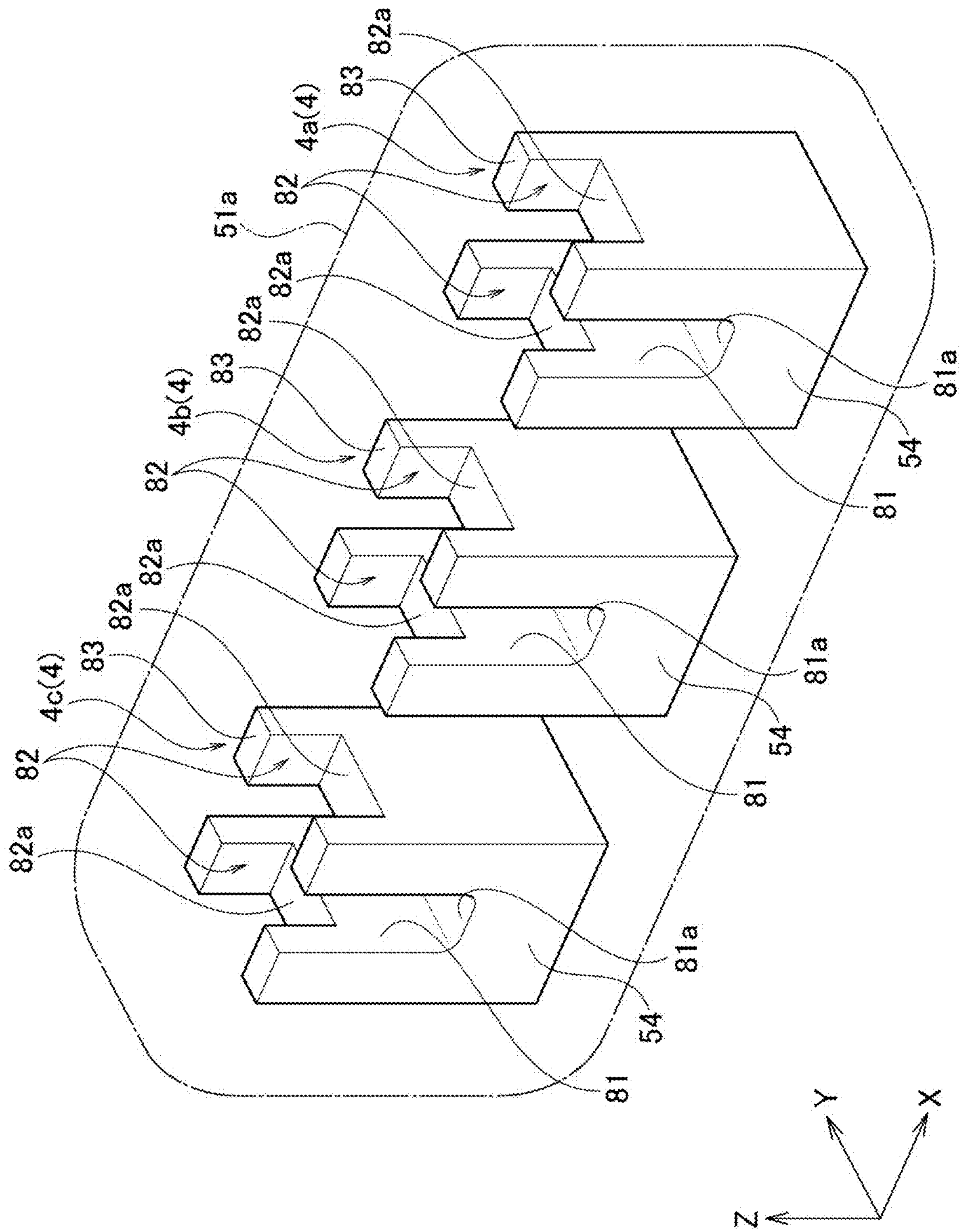
60
2
Y
X
Z

FIG. 1

FIG. 2

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83
4a(4)
82
51a
82a
82a
83
4b(4)
82
82a
82a
83
4c(4)
82
82a
81a
54
81
81a
54
81
81a
54
81
Y
X
Z

FIG. 3

FIG. 4

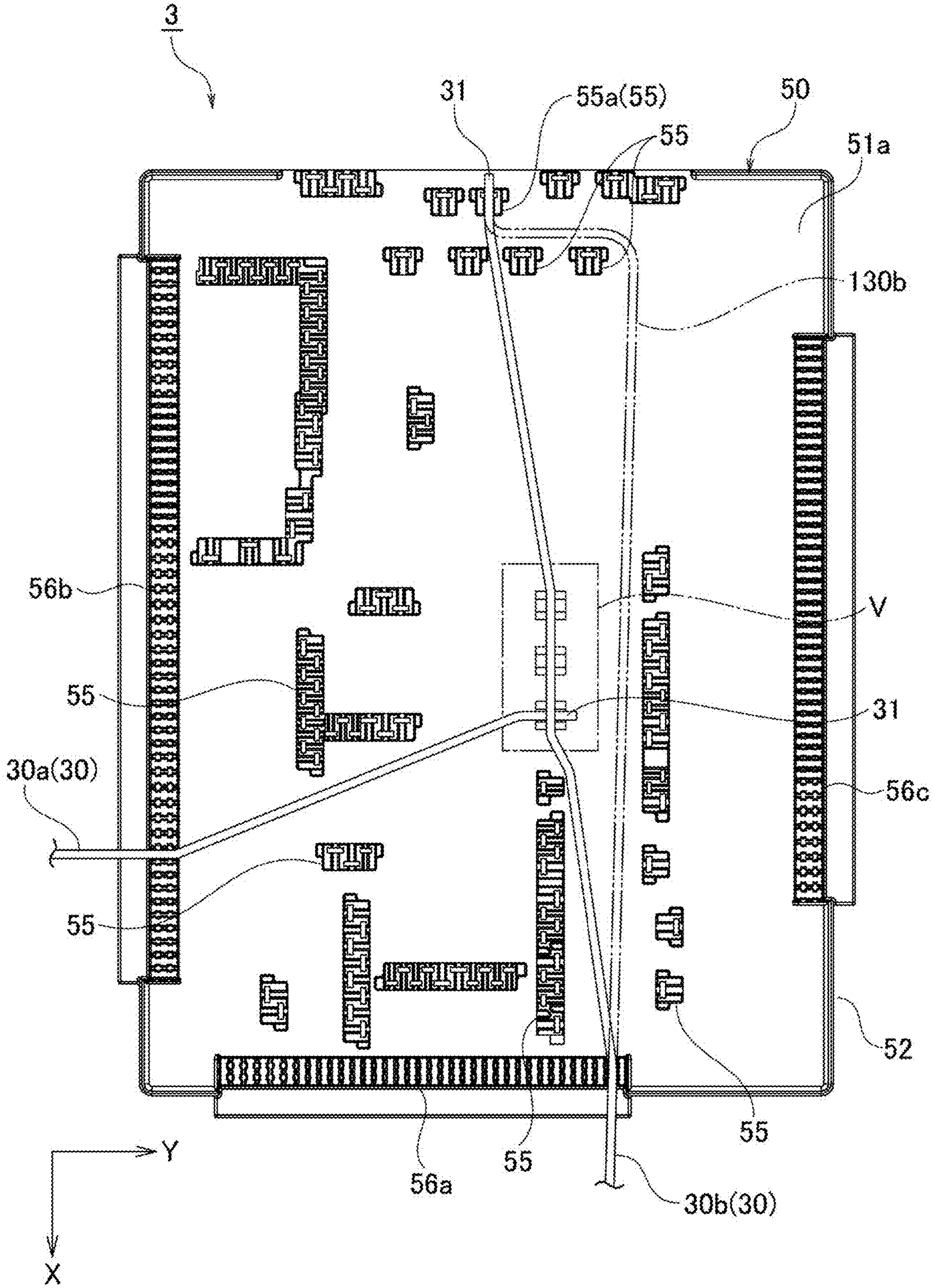
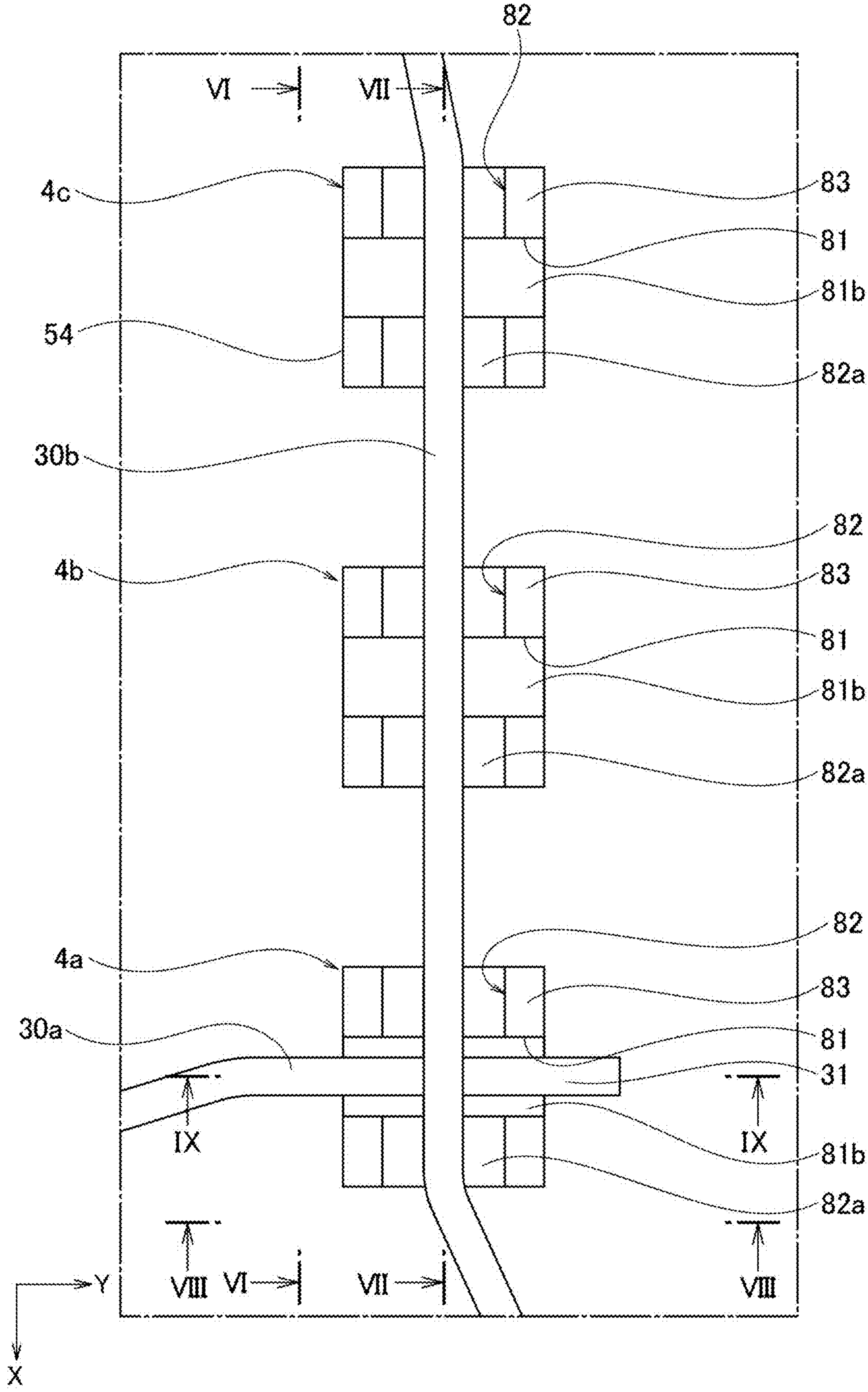

3
31
55a(55)
55
50
51a
130b
56b
V
55
31
30a(30)
56c
55
52
55
55
56a
30b(30)
Y
X

FIG. 5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VI
VII
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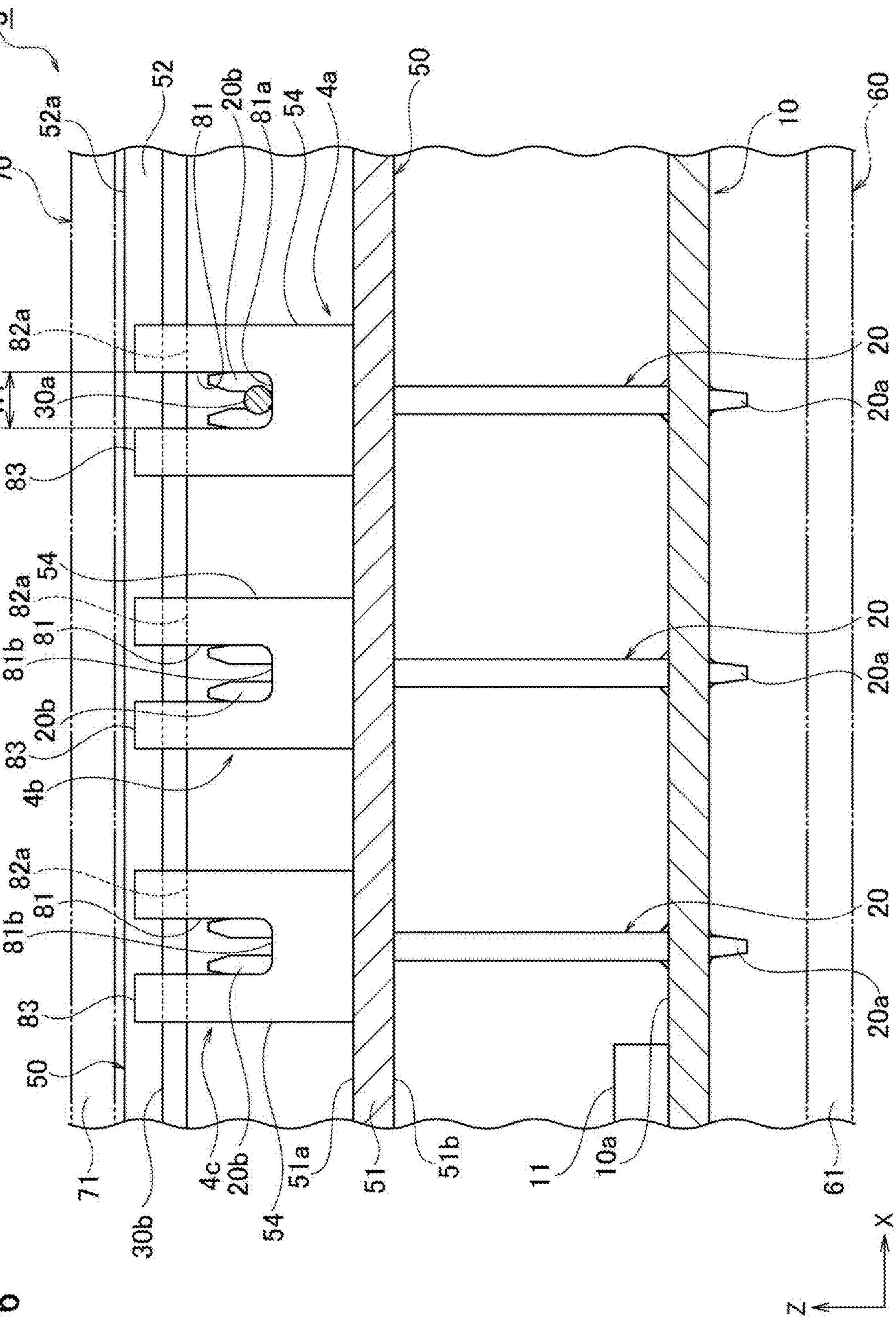
83
81
81b
54
82a
30b
82
4b
83
81
81b
82a
82
4a
83
30a
81
31
IX
IX
81b
82a
VIII
VI
VII
VIII
Y
X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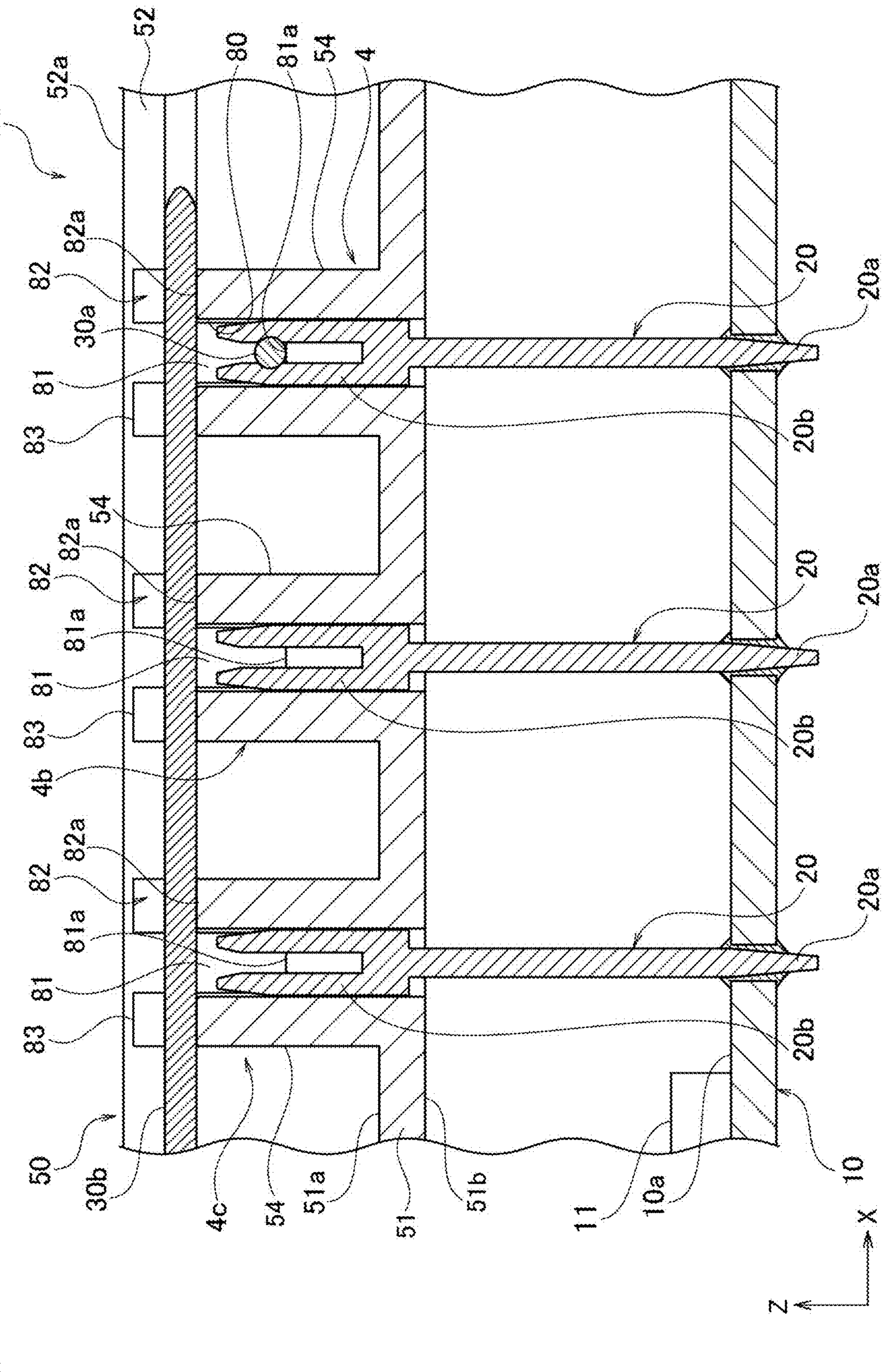
70
52a
52
81
20b
81a
54
4a
50
60
10
82a
W1
30a
83
20
20a
54
82a
81b
81
20b
83
4b
81b
82a
81
83
50
71
30b
4c
20b
54
51a
51
51b
11
10a
61
X
Z

FIG. 6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52a
52
80
81a
54
4
82a
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30a
81
83
20
20a
20b
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82a
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81a
81
83
4b
82a
82
81a
81
83
50
30b
4c
54
51a
51
51b
11
10a
10
X
Z

FIG. 7

FIG. 8

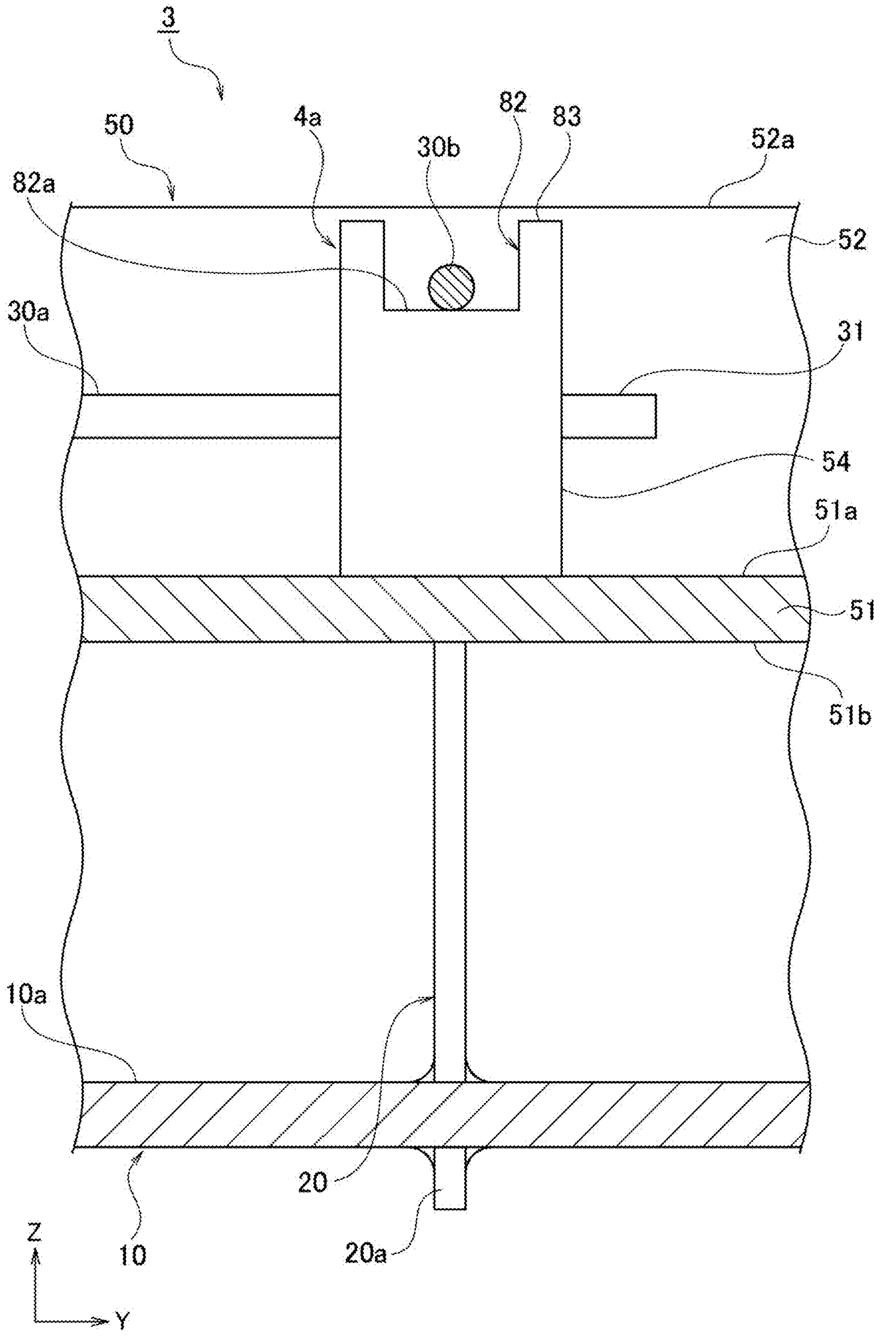


FIG. 9

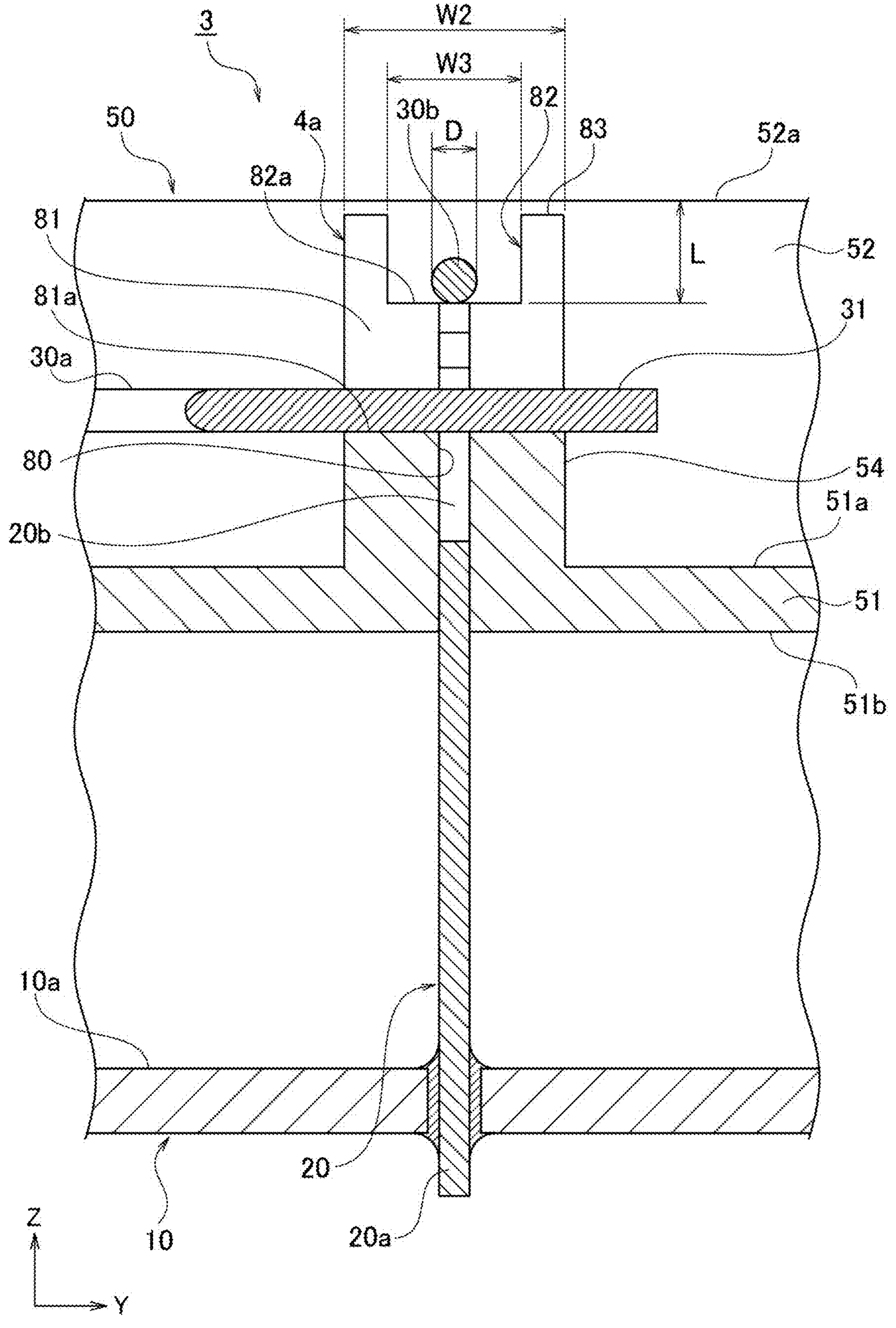

3
W2
W3
82
30b
D
83
50
4a
52a
82a
81
L
52
81a
31
30a
80
54
51a
20b
51
51b
10a
20
10
20a
Z
Y

FIG. 10

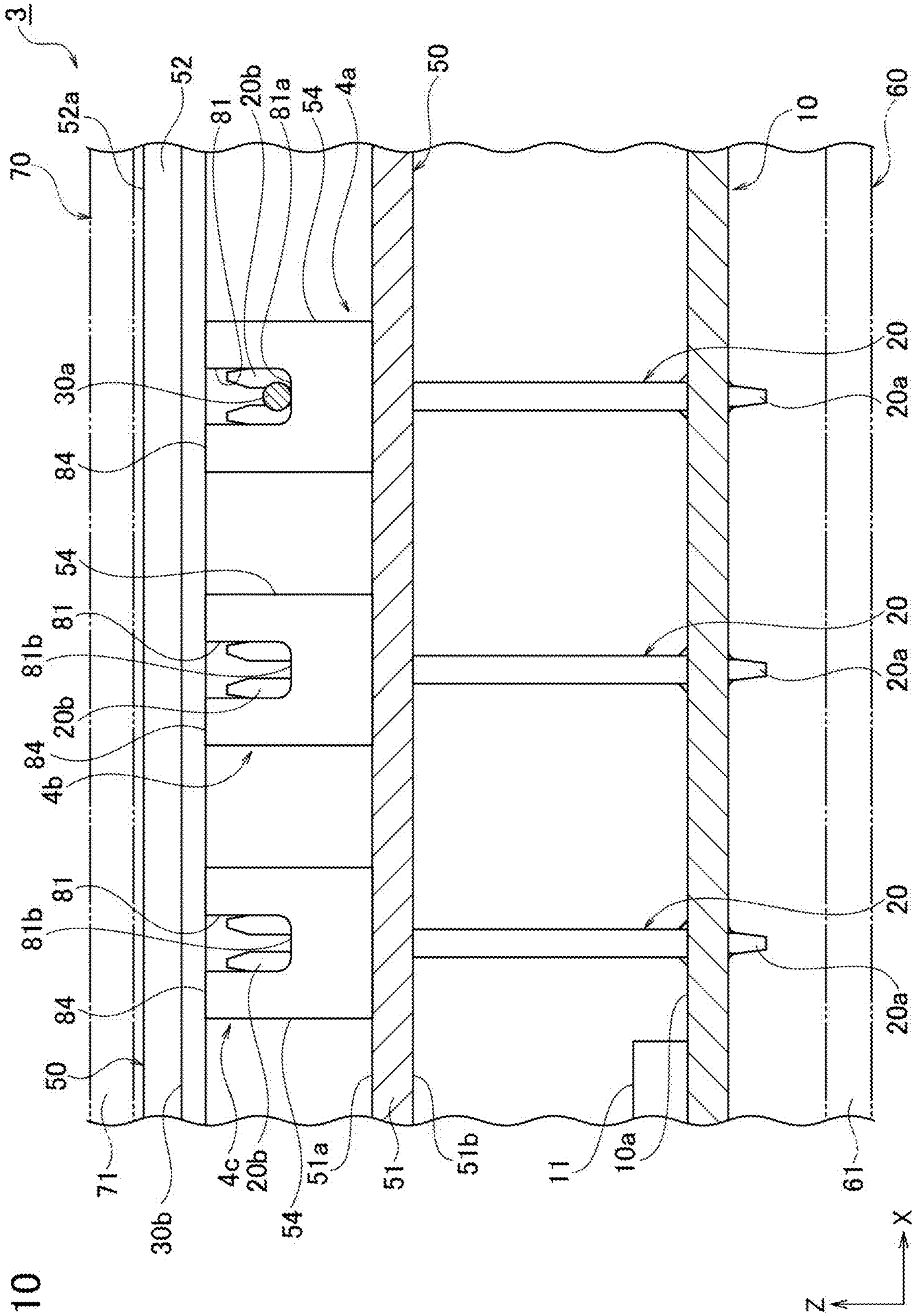
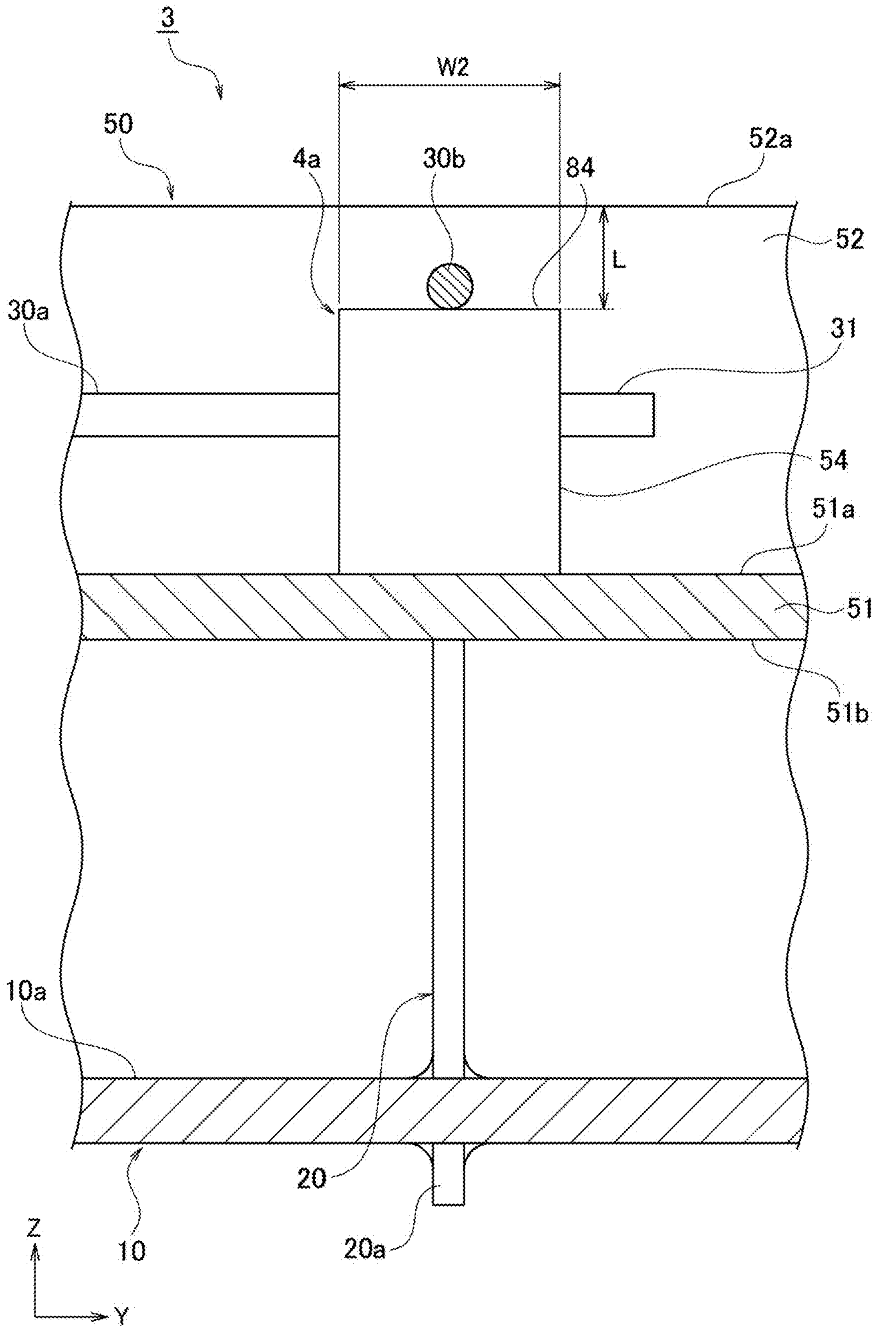

3
70
52a
52
81
20b
81a
54
4a
50
10
60
30a
84
54
81
81b
20b
84
4b
20
20a
20
20a
81b
81
84
50
20
20a
71
30b
4c
20b
54
51a
51
51b
11
10a
61
X
Z

FIG. 11



ELECTRONIC CONTROL UNIT

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application is based on, and claims priority from the Japanese Patent Application No. 2024-036602, filed on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an electronic control unit.

BACKGROUND

[0003] Conventionally, vehicles such as automobiles are equipped with an electronic control unit used for controlling various functions. In general, the electronic control unit includes a substrate on which electronic components are mounted. As an example of such a substrate, JP 2002-58129 A discloses a technology related to an electrical connection box including a wiring board having a plurality of reinforcing projections for supporting male terminals for fuse connections.

SUMMARY OF THE INVENTION

[0004] On a substrate such as a wiring board disclosed in JP 2002-58129 A, a configuration in which an electric wire is connected to a terminal having one end joined to the substrate can also be considered. When a plurality of electric wires are routed on a substrate, a routing board arranged in parallel with the substrate is further provided, a plurality of electric wires are routed on the routing board, and end portions of the electric wires may be connected to a part of a terminal exposed on the routing board in advance. However, when, for example, a plurality of terminals exposed on the routing board are randomly arranged, and a large number of electric wires are intermingled on a routing board, there is room for improvement in order to increases degrees of freedom for wiring.

[0005] An object of the present disclosure is to provide an electronic control unit which increases degrees of freedom for wiring.

[0006] An electronic control unit according to an aspect of an embodiment includes a plurality of terminals; a plurality of electric wires having a first end portion connected to one end of the plurality of terminals; and a routing board for routing the plurality of electric wires introduced from outside, the routing board including a base having a routing surface facing the plurality of electric wires that are routed; a side wall provided on an edge of the base and projecting in a normal direction of the routing surface; and a plurality of terminal blocks provided on the routing surface; in which the terminal blocks have an opening groove, which is penetrating in a first direction orthogonal to the normal direction of the routing surface, and inside of which a part of the end of the terminals is exposed; at least a part of an upper surface of at least one of the terminal blocks is a routing region surface crossing in a second direction orthogonal to the normal direction of the routing surface and the first direction; a height position of the routing region surface from the routing surface is set higher than a bottom surface of the opening groove, and lower than a height position of a tip edge of the side wall by a distance that is larger than a dimension value of an outer diameter of the electric wires, and a width of the routing region surface in the first direction is set larger than the dimension value of the outer diameter of the electric wires.

[0007] According to the configuration above, it is possible to provide an electronic control unit, inside of which degrees of freedom for wiring are increased.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a perspective view of an electronic control unit according to one embodiment.

[0009] FIG. **2** is a perspective view of a circuit unit included in the electronic control unit.

[0010] FIG. **3** is an enlarged perspective view of electric wire connections corresponding to part III of FIG. **2**.

[0011] FIG. **4** is a plan view of a circuit unit including an example of wiring on a routing board.

[0012] FIG. **5** is an enlarged plan view of the electric wire connections corresponding to part V in FIG. **4**.

[0013] FIG. **6** is a side view of the electric wire connections corresponding to a cross section VI-VI in FIG. **5**.

[0014] FIG. **7** is a cross-sectional view of the electric wire connections corresponding to a cross section VII-VII in FIG. **5**.

[0015] FIG. **8** is a side view of the electric wire connections corresponding to a cross section VIII-VIII in FIG. **5**.

[0016] FIG. **9** is a cross-sectional view of the electric wire connection corresponding to a cross section IX-IX in FIG. **5**.

[0017] FIG. **10** is a side view of another example of the wire connections illustrated in accordance with FIG. **6**.

[0018] FIG. **11** is a side view of another example of the wire connections illustrated in accordance with FIG. **8**.

DETAILED DESCRIPTION OF THE INVENTION

[0019] Hereinafter, an electronic control unit according to one embodiment will be described in detail with reference to the drawings. Hereinafter, the term "electronic control unit" will be abbreviated as "ECU." The dimensional ratios in the drawings are exaggerated for the sake of explanation and may differ from the actual ratios.

[0020] FIG. **1** is a perspective view of an ECU **1** according to one embodiment. FIG. **2** is a perspective view of a circuit unit **3** included in the ECU **1**.

[0021] The ECU **1** is mounted in a vehicle such as an automobile, and controls various functions. The ECU **1** includes a substrate **10**, a plurality of terminals **20**, a plurality of electric wires **30**, a connector **40**, a routing board **50**, a first cover **60**, and a second cover **70**.

[0022] Here, in the present embodiment, a unit including the substrate **10**, the plurality of terminals **20**, a part of the plurality of electric wires **30**, the routing board **50**, the first cover **60**, and the second cover **70** is denoted as "main body **2**." A unit consisting of the substrate **10**, the plurality of terminals **20**, the plurality of electric wires **30**, the connector **40**, and the routing board **50** is denoted as "circuit unit **3**."

[0023] The substrate **10** is, for example, a printed circuit board (PCB) on which a plurality of electronic components **11** (see FIG. **6**, for example.) are mounted in advance. Examples of the plurality of electronic components **11** include a CPU (Central Processing Unit) and a memory. In the present embodiment, the substrate **10** is regarded as a substantially rectangular flat plate which is parallel to a virtual XY plane defined by an X direction and a Y direction orthogonal to each other, and which has two ends in the X direction and two ends in the Y direction.

[0024] The terminals **20** electrically connect one of the electric wires **30** to a predetermined circuit on the substrate **10**. The terminals **20** are a flat, rod-shaped metal member. Each of the terminals **20** has a first end **20***a* which is one end, and a second end **20***b* which is the other end opposite to the first end **20***a* (see FIG. **6**, for example). The first end **20***a* is connected to a predetermined circuit on the substrate **10**. In the present embodiment, since the substrate **10** is regarded as a flat plate parallel to the XY plane, the terminals **20** are each arranged on a mounting surface **10***a* (see FIG. **6**, for example) so as to project in a Z direction orthogonal to both the X and Y directions. The second end **20***b* connects a first end portion **31** (see FIG. **4**) of one of the electric wires **30**. The second end **20***b* includes, for example, two protruding pieces parallel to each other, and a core wire of the electric wires **30** is clamped between the two protruding pieces, so that the first end portion **31** can be easily connected. At least a part of the second end **20***b* is supported by a first terminal block **54** or a second terminal block **55**. At least a part of the second end **20***b* supported by the first terminal block **54** or the like is exposed through an opening groove **81** on a routing surface **51***a* of a base **51**.

[0025] The electric wires **30** each electrically connect the connector **40** and one of the terminals **20**. The electric wires **30** are electric wires in which a core wire as a conductor, is coated with an insulator. Each of the electric wires **30** has the first end portion **31** which is one end, and a second end portion **32** which is the other end opposite to the first end portion **31**. As described above, the first end portion **31** is connected to the second end **20***b* of the terminals **20**. The second end portion **32** is connected to the connector **40**. In both of the first end portion **31** and the second end portion **32**, a part that actually comes into contact with the second end **20***b* or the connector **40** is a core wire portion partially exposed by removing the insulator.

[0026] The connector **40** is connected to an external connector (not illustrated) that is provided to an object to which the ECU **1** is attached. In the present embodiment, the ECU **1** includes, as an example, three connectors **40**, that is, a first connector **40***a*, a second connector **40***b*, and a third connector **40***c*. The second end portion **32** of each of the electric wires **30** is connected to one of the first connector **40***a*, the second connector **40***b*, and the third connector **40***c*. The connectors **40** are not directly provided on the substrate **10**, but are connected to the main body **2** of the ECU 1 only through the electric wire wires **30**.

[0027] The routing board **50** routes each of the electric wires **30** while connecting the first end portion **31** to the terminals **20** corresponding to each of the electric wires **30** inside the main body **2**. The routing board **50** is made of, for example, a synthetic resin, and is a flat member arranged in parallel with the substrate **10**. Since the routing board **50** is maintained in a stacked position with respect to the substrate **10**, it may also be called a "wiring layer." The routing board **50** has the base **51**, a side wall **52**, a pedestal **53**, at least one first terminal block **54** (see FIG. **6**, for example.), a plurality of second terminal blocks **55**, and electric wire inlets **56**.

[0028] The base **51** is a plate facing the substrate **10** in the Z direction. The planar shape of the base **51** is equivalent to that of the substrate **10**. One surface of the base **51** is the routing surface **51***a* on which a part of each of the electric wires **30** is wired. A normal direction of the routing surface **51***a* is in the Z direction. The other surface of the base **51** is on a side opposite to the routing surface **51***a,* and is a back surface **51***b* facing the substrate **10** (See FIG. **6**, for example.).

[0029] The side wall **52** is continuous with an edge of the base **51**, and protrudes in the Z direction, that is, in the normal direction of the routing surface **51***a,* to surround the routing space on the routing surface **51***a*. However, the side wall **52** is not in the form of a frame surrounding entire four sides in the present embodiment, but is continuous with at least one of the electric wire inlets **56**.

[0030] The pedestal **53** is continuous with the edge of the base **51**, and protrudes in the Z direction to be at least partly in contact with an outer edge of the mounting surface **10***a* of the substrate **10**. Thus, a constant distance between the base **51** and the substrate **10** is maintained.

[0031] The first terminal blocks **54** and the second terminal blocks **55** are provided on the routing surface **51***a,* and support the terminals **20** with the second end **20***b* thereof being exposed. The arrangement of the terminals **20** depends on the circuit configuration of the mounting surface **10***a* of the substrate **10**. Therefore, the arrangement of the first terminal blocks **54** and the second terminal blocks **55** is also determined based on the circuit configuration of the substrate **10**.

[0032] FIG. **3** is an enlarged perspective view of electric wire connections **4** corresponding to part III of FIG. **2**. Here, the electric wire connections **4** refer to components in which the electric wires **30** are connected to the second end **20***b* of the terminals **20** in the first terminal blocks **54** or the second terminal blocks **55**.

[0033] The first terminal blocks **54** are terminal blocks having a routing area surface **82***a* on at least a part of a first upper surface **83**. In the present embodiment, as an example, it is assumed that three terminals **20** are present as the first terminals, and in addition. The first terminal blocks **54** have an opening groove **81**, inside of which a part of the second end **20***b* is exposed. For each of the electric wires **30**, a part of the first end portion **31** is inserted into the opening groove **81** that is predetermined, so that the core wire of the first end portion **31** is connected to the second end **20***b*. Hereinafter, the three electric wire connections **4** including the first terminal blocks **54** are referred to as a first electric wire connection **4***a*, a second electric wire connection **4***b*, and a third wire connection **4***c*. The first electric wire connection **4***a*, the second electric wire connection **4***b*, and the third wire connection **4***c* are lined up in a direction opposite to the X direction. The specific configuration of the electric wire connections **4** including the first terminal blocks **54** will be described in detail below.

[0034] The second terminal blocks **55** are terminal blocks with no surface, such as the routing region surface **82***a,* on an upper surface. In other words, the shape of the second terminal blocks **55** is the same as that of the first terminal blocks **54** except that it does not have an electric wire passage groove **82** including the routing region surface **82***a* as a bottom surface. It should be noted that the second terminal blocks **55** may have a shape in which a plurality of second terminal blocks **55** supporting the second end of one of the terminals **20** are integrated in parallel and continuously.

[0035] In this embodiment, among the plurality of terminal blocks on the routing surface **51***a,* three first terminal blocks **54** are present, and terminal blocks other than the first terminal blocks **54** are the second terminal blocks **55**. However, among the plurality of terminal blocks, the number of the first terminal blocks **54** may be larger than the number of the second terminal blocks**55**. Furthermore, all of the plurality of terminal blocks may be the first terminal blocks **54**. In other words, the presence of the second terminal blocks **55** is not essential.

[0036] The electric wire inlets **56** introduce the electric wires **30** onto the routing surface **51***a* of the base **51** from the outside of the routing board **50** while supporting a part of the electric wires **30**. At least one of the electric wire inlets **56** is provided as a convex portion at an outer edge of the routing board **50** and on the routing surface **51***a*. In the present embodiment, as an example, three electric wire inlets **56** are present: a first electric wire inlet **56***a* extending in the Y direction, and a second electric wire inlet **56***b* and a third electric wire inlet **56***c*, each extending in the X direction. When routing the electric wires **30**, not necessarily all of the electric wire inlets **56** are necessarily to be used. FIG. **1** illustrates, as an example, a configuration in which all the electric wires **30** are pulled into the body **2** from the connector **40** only through the first electric wire inlet **56***a*.

[0037] The first cover **60** and the second cover **70** are combined with each other to accommodate at least the substrate **10**, the plurality of terminals **20**, and the routing board **50** included in the circuit unit **3**. The first cover **60** and the second cover **70** may be made of a synthetic resin or metal, for example.

[0038] The first cover **60** is a cover to be assembled from a side facing the substrate **10**. The first cover **60** has a first flat plate **61** (see FIG. **6**) covering the substrate **10** and a first frame **62** continuous with an edge of the first flat plate **61**. The first frame **62** projects in the Z direction in which is the first frame **62** is assembled with respect to the second cover **70**, so as to surround the substrate **10**, for example. A part of a first tip edge **62***a* of the first frame **62** is close to the electric wire inlets **56** of the routing board **50** in a non-contact manner.

[0039] The second cover **70** is a cover to be assembled from a side facing the routing surface **51***a* of the routing board **50**. The second cover **70** has a second flat plate **71** (see FIG. **6**) covering the routing board **50**, and a second frame **72** continuous with an edge of the second flat plate **71**. The second frame **72** projects in a direction opposite to the Z direction, which is the direction to be assembled with respect to the first cover **60**, so as to surround at least a part of the routing board **50**, for example. A part of a second tip edge **72***a* of the second frame **72** is close to the electric wire inlets **56** of the routing board **50** in a non-contact manner. A part of an area between the first tip edge **62***a* of the first frame **62** of the first cover **60** and the second tip edge **72***a* of the second frame **72** of the second cover **70** is an open area through which the plurality of electric wires **30** pass inside and outside the main body **2**.

[0040] The first cover **60** has a plurality of first locking portions **63** in the first frame **62**. On the other hand, the second cover **70** has a plurality of second locking portions **73** in the second frame **72**. The second cover **70** is combined with the first cover **60** by engaging the second locking portions **73** with the first locking portions **63**.

[0041] Next, a specific configuration of the electric wire connections **4** including the first terminal blocks **54** will be described.

[0042] FIG. **4** is a plan view of the circuit unit **3** including an example of wiring on the routing board **50**. FIG. **5** is an enlarged plan view of the electric wire connections **4** including the first terminal blocks **54** corresponding to part V in FIG. **4**.

[0043] Here, although a large number of the electric wires **30** can actually be wired in the routing board **50**, only two electric wires **30**, the first electric wire **30***a* and the second electric wire **30***b*, will be exemplified below for the sake of simplicity.

[0044] The first electric wire **30***a* is an example of the electric wires **30** pulled into the main body **2** from a connector (not illustrated) similar to the connector **40** through the second electric wire inlet **56***b*. The first electric wire **30***a* is linearly wired from the second electric wire inlet **56***b* toward the first electric wire connection **4***a* on the routing surface **51***a*, for example, passing through gaps in the plurality of second terminal blocks **55**. The first end portion **31** of the first electric wire **30***a* is connected to the second end **20***b* of the terminals **20** located in the first electric wire connection **4***a*.

[0045] The second electric wire **30***b* is an example of the electric wires **30** pulled into the main body **2** from the connector **40** through the first electric wire inlet **56***a*. The second electric wire **30***b* is linearly routed from the first electric wire inlet **56***a* to the second end **20***b* of the terminals **20** supported by one of the second terminal blocks **55**, for example, passing through gaps in the plurality of second terminal blocks **55** on the routing surface **51***a*. Hereinafter, the second terminal block **55** supporting the second end **20***b* to which the first end portion **31** of the second electric wire **30***b* is connected is referred to as a "connection terminal block **55***a*" for convenience. The connection terminal block **55***a* is provided in the vicinity of another side wall **52** opposed in the X direction to the side wall **52** in which the first electric wire inlet **56***a* is provided.

[0046] The second electric wire **30***b* passes through the electric wire passage groove **82** formed in the first terminal block **54** included in each of the first electric wire connection **4***a*, the second electric wire connection **4***b*, and the third electric wire connection **4***c*. The second electric wire **30***b* passes above the first electric wire **30***a* connected to the first electric wire connection **4***a*. Here, the "above" with respect to each member refers to an area corresponding to an upper side when the routing surface **51***a* is a horizontal plane, and the normal direction of the routing surface **51***a* is from bottom to top.

[0047] FIGS. **6** and **7** are views of the first electric wire connection **4***a*, the second electric wire connection **4***b*, and the third electric wire connection **4***c* as viewed in the Y direction. FIG. **6** is a side view of the electric wire connections **4** corresponding to a cross section VI-VI in FIG. **5**. FIG. **7** is a cross-sectional view of the electric wire connections **4** corresponding to a cross section VII-VII in FIG. **5**. The cross section in FIG. **7** is in an XZ plane passing through an intermediate position in a thickness direction corresponding to the Y direction of the terminals **20** supported by the first terminal blocks **54**.

[0048] FIGS. **8** and **9** are views of the first electric wire connection **4***a* viewed in the negative X direction. FIG. **8** is a side view of the first electric wire connection **4***a* corresponding to a cross section VIII-VIII in FIG. **5**. FIG. **9** is a cross-sectional view of the first electric wire connection **4***a* corresponding to a cross section IX-IX in FIG. **5**. The cross section in FIG. **9** is in the YZ plane passing through an intermediate position in a width direction corresponding to the X direction of the terminals **20** supported by the first terminal blocks **54**.

[0049] The three electric wire connections **4** including the first terminal blocks **54** have the same configuration. Therefore, the first electric wire connection **4***a* will be mainly described with respect to the shape or the like.

[0050] First, as described above, the second ends **20***b* of the terminals **20** are supported by the first terminal blocks **54** so that at least a part thereof is exposed on the routing surface **51***a* , thereby connecting the electric wires **30** by clamping the core wires of the electric wires **30** between the two protruding pieces above the routing surface **51***a*.

[0051] The first terminal blocks **54** are provided for each of the terminals **20** in the present embodiment, and therefore, three first terminal blocks **54** are present for the first electric wire connection **4***a*, the second electric wire connection **4***b*, and the third electric wire connection **4***c*. As in the example of the second terminal blocks **55** described above, the first terminal blocks **54** may have a shape such that the three first terminal blocks **54** supporting the second end **20***b* of each of the terminals **20** are integrated in parallel and continuously.

[0052] The overall schematic shape of the first terminal blocks **54** is a rectangular parallelepiped whose one end is continuous with the routing surface **51***a* of the base **51**. Hereinafter, an upper surface of the first terminal block **54** refers to a surface of the first terminal block **54** that is located on an upper side in the normal direction of the routing surface **51***a* corresponding to the Z direction, and faces the second flat plate **71** of the second cover **70** in the Z direction when the ECU 1 is assembled. When the ECU 1 is assembled, the first upper surfaces **83** face the second flat plate **71** of the second cover **70** in the Z direction, as illustrated in FIG. **6**. The first terminal blocks **54** have a terminal through-hole **80**, an opening groove **81**, and an electric wire passage groove **82**.

[0053] The terminal through-hole **80** accommodates the second end **20***b* of the terminals **20**. The terminal through-hole **80** is penetrating in the Z direction in which the base **51** faces the substrate **10**. One opening of the terminal through-hole **80** is located on a bottom surface **81***a* of the opening groove **81**. The other opening of the terminal through-hole **80** is located on the back surface **51***b* of the base **51** in this embodiment. In this embodiment, the direction (hereinafter referred to as "clamping direction") in which the second end **20***b* clamps the first electric wire **30***a* is in the X direction. Therefore, in order to stably support the second end **20***b*, a part of walls defining the terminal through-hole **80** may be a pair of fitting groove portions which face each other in the X direction in accordance with the clamping direction to fit a part of the second end **20***b*.

[0054] The opening groove **81** is a groove which is opened at the first upper surface **83** and is penetrating in the first direction orthogonal to the normal direction of the routing surface **51***a*. In this embodiment, the first direction corresponds to the Y direction. The opening groove **81** communicates with the terminal through-hole **80** and exposes a part of the second end **20***b* of the terminals **20** inside. In the opening groove **81**, it is necessary to satisfy the condition that the first end portion **31** of the first electric wire **30***a* is introduced in the first direction, and the second end **20***b* clamps and connects the first end portion **31** that has been introduced. Therefore, a first width W**1** of the opening groove **81** in the second direction along the clamping direction is set larger than a dimension value of the outer diameter D of the first electric wire **30***a*, as illustrated in FIG. **6**. On the other hand, a second width W**2** in the first direction orthogonal to the first width W**1** in the opening groove **81** is the same as the width of the first terminal block **54** in the first direction.

[0055] The opening groove **81** has the bottom surface **81***a* which is separated from the height position of the first upper surface **83** in the direction opposite to the normal direction of the routing surface **51***a*, and is separated from the height position of the routing surface **51***a* in the normal direction. The bottom surface **81***a* is substantially parallel to the first upper surface **83** and the routing surface **51***a*. The height position of the bottom surface **81***a* may be set so that the second end **20***b* of the terminals **20** connects the first end portion **31** at a desired position when the first end portion **31** of the first electric wire **30***a* introduced into the opening groove **81** is in contact with the bottom surface **81***a*.

[0056] The electric wire passage groove **82** is a groove which is opened at the first upper surface **83**, and is penetrating in the second direction orthogonal to the normal direction of the routing surface **51***a*. In this embodiment, the second direction corresponds to the X direction. The electric wire passage groove **82** allows a second electric wire **30***b* to pass, which is different from the first electric wire **30***a* connected to the terminals **20**, supported by the first terminal block **54**. Therefore, the electric wire passage groove **82** is formed to cross a part of the opening groove **81** in the first terminal block **54**.

[0057] Further, the electric wire passage groove **82** has, as a bottom surface thereof, the routing region surface **82***a* which crosses the first terminal block **54** in the second direction. The routing region surface **82***a* is an end surface located opposite to a virtual end surface P in the Z direction, and it can be regarded as a part of the upper surface of the first terminal block **54**.

[0058] The height position of the routing region surface **82***a* in the Z direction with respect to the routing surface **51***a* is set higher than the bottom surface **81***a* of the opening groove **81**. Further, the height position of the routing region surface **82***a* is set to a position lower than the height position of a tip edge **52***a* of the side wall **52**, by a distance L, as illustrated in FIG. **9**. The distance L is larger than the dimension value of the outer diameter D of the electric wires **30**.

[0059] A third width W**3** in the first direction of the routing region surface **82***a* is set to be narrower than the second width W**2**, and larger than the dimension value of the outer diameter D of the electric wires **30**, as illustrated in FIG. **9**. Here, the third width W**3** may have a dimension value more than twice the outer diameter D of the electric wires **30** so that the plurality of electric wires **30** can pass through the electric wire passage groove **82** at the same time.

[0060] Next, effects of the ECU **1** will be described.

[0061] The ECU **1** includes the plurality of terminals **20**; the plurality of electric wires **30** having the first end portion **31** connected to one end of the plurality of terminals **20**; and the routing board **50** for routing the plurality of electric wires **30** introduced from outside. The routing board **50** has the base **51** having the routing surface **51***a* facing the plurality of electric wires **30** that are routed; the side wall **52** provided on an edge of the base **51** and projecting in a normal direction of the routing surface **51***a;* and a plurality of terminal blocks provided on the routing surface **51***a*. The terminal blocks have an opening groove **81**, which is penetrating in a first direction orthogonal to the normal direction of the routing surface **51***a,* and inside of which a part of the second end **20***b* of the terminals **20** is exposed. At least a part of an upper surface of terminal block **54**, which is at least one of the terminal blocks, is a routing region surface crossing in a second direction orthogonal to the normal direction of the routing surface **51***a* and the first direction. The height position of the routing region surface from the routing surface **51***a* is set higher than the bottom surface **81***a* of the opening groove **81**, and lower than the dimension value of the outer diameter D of the electric wires **30** from the height position of the tip edge **52***a* of the side wall **52**. The third width W3 of the routing region surface in the first direction is set larger than the dimension value of the outer diameter D of the electric wires **30**.

[0062] In addition, in the ECU 1, the first terminal block **54** may have the electric wire passage groove **82** allowing, for example, the second electric wire **30***b* to pass, which is different from, for example, the first electric wire **30***a* connected to the terminal **20** supported by the first terminal block **54**. The routing region surface may be the routing region surface **82***a* which is a bottom surface of the electric wire passage groove **82**.

[0063] In the example above, the normal direction of the routing surface **51***a* corresponds to the Z direction, the first direction corresponds to the Y direction, and the second direction corresponds to the X direction.

[0064] Here, as a comparative example, it is assumed that all of the first electric wire connection **4***a,* the second electric wire connection **4***b,* and the third electric wire connection **4***c,* which include the first terminal block **54** in the example above, includes the second terminal block **55** without the electric wire passage groove **82**, instead of the first terminal block **54**. According to the comparative example, the first electric wire **30***a* can be wired in the same wiring path as in the example above. However, the second terminal block **55** included in the first electric wire connection **4***a* does not have the electric wire passage groove **82**, and closely faces the second flat plate **71** of the second cover **70** with an upper surface, so that the second electric wire **30***b* cannot pass above the second terminal block **55**. Therefore, as illustrated in FIG. **4**, a virtual electric wire **130***b* in place of the second electric wire **30***b* is routed on a wiring path that avoids a plurality of the second terminal blocks **55**. In other words, the wiring path of the virtual electric wire **130***b* is assumed to be more complicated than that of the second electric wire **30***b*. Moreover, the length of the virtual electric wire **130***b* is assumed to be longer than that of the second electric wire **30***b*.

[0065] In the ECU 1 according to the present embodiment, all of the first electric wire connection **4***a,* the second electric wire connection **4***b,* and the third electric wire connection **4***c* includes the first terminal block **54** having the electric wire passage groove **82** provided to cross the opening groove **81**. The height position of the routing region surface **82***a,* which is a bottom surface of the electric wire passage groove **82**, is higher than the bottom surface **81***a* of the opening groove **81**, and lower than the dimension value of the outer diameter D of the electric wires **30** from the height position of the tip edge **52***a* of the side wall **52**. In addition, the third width W3 of the routing region surface **82***a* satisfies the condition that it is wider than the dimension value of the outer diameter D of the electric wires **30**. Therefore, for example, if it is desired to route some of the electric wires **30**, such as the second electric wire **30***b,* in a simple linear path, the electric wire passage groove **82** can be used as a routing region of the second electric wire **30***b,* so that interference with the first terminal blocks **54** is difficult to occur. Therefore, according to the ECU 1, the wiring freedom of the electric wires **30** can be increased. In particular, instead of the examples illustrated in the drawings, the wiring freedom of the electric wires **30** can be further increased by using the first terminal blocks **54** for all the terminal blocks provided on the routing surface **51***a*.

[0066] As described above, according to the present embodiment, it is possible to provide the ECU 1, inside of which the degrees of freedom for wiring are increased.

[0067] According to the ECU 1, the wiring path is simplified for each of the electric wires **30** by increasing the degrees of freedom for wiring, wire saving of the plurality of electric wires **30** is possible, and the main body **2** can be miniaturized or reduced in weight by increasing the wiring space on the routing board **50**.

[0068] Furthermore, in the ECU 1, the electric wires **30** such as the second electric wire **30***b* passing through the electric wire passage groove **82** are prevent from moving in the first direction corresponding to a penetrating direction of the opening groove **81** by the electric wire passage groove **82**. Therefore, according to the ECU 1, it is advantageous that the wiring paths of the plurality of electric wires **30** can be easily maintained in a desired state in the routing board **50**.

[0069] Here, in the examples above with reference to the drawings, the routing region surface is assumed to be the routing region surface **82***a* which is the bottom surface of the electric wire passage groove **82**. However, the embodiments are not limited thereto, and an entire top surface of the first terminal blocks **54** may be used as a routing region surface.

[0070] FIG. **10** is a side view of another example of the electric wire connections **4** illustrated in accordance with FIG. **6**. FIG. **11** is a side view of another example of the electric wire connections **4** illustrated in accordance with FIG. **8**.

[0071] In other examples of the electric wire connections **4** illustrated in FIGS. **10** and **11**, the upper surface of the first terminal blocks **54** is a second upper surface **84** instead of the first upper surface **83**. The height position of the second upper surface **84** in the normal direction with respect to the routing surface **51***a* is set to the height position of the routing area surface **82***a* when the first terminal block **54** has the electric wire passage groove **82**. Therefore, according to the ECU 1 as described above, the electric wires **30** such as the second electric wire **30***b* can pass above the first terminal block **54** in the same way as the case including the first terminal block **54** having the electric wire passage groove **82**, so that the degrees of freedom of wiring inside the ECU 1 can be increased.

[0072] Furthermore, the ECU 1 may include the connectors **40** to which the second end portion **32** opposite to the first end portion **31** of each of the electric wires **30** is connected.

[0073] In this ECU 1, the connectors **40** are only connected to the main body **2** via the plurality of electric wires **30**, so that positions of the connectors **40** can be freely changed relative to the main body **2**. Therefore, according to the ECU 1, it is possible to avoid, for example, the occurrence of troubles such as reduced space in a passenger compartment due to bulging of external electric wires around the main body **2**.

[0074] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions, and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

1. An electronic control unit comprising:

a plurality of terminals;

a plurality of electric wires having a first end portion connected to one end of the plurality of terminals; and

a routing board for routing the plurality of electric wires introduced from outside;

the routing board comprising:

a base having a routing surface facing the plurality of electric wires that are routed;

a side wall provided on an edge of the base and projecting in a normal direction of the routing surface; and

a plurality of terminal blocks provided on the routing surface;

wherein

the terminal blocks have an opening groove, which is penetrating in a first direction orthogonal to the normal direction of the routing surface, and inside of which a part of the end of the terminals is exposed;

at least a part of an upper surface of at least one of the terminal blocks is a routing region surface crossing in a second direction orthogonal to the normal direction of the routing surface and the first direction;

a height position of the routing region surface from the routing surface is set higher than a bottom surface of the opening groove, and lower than a height position of a tip edge of the side wall by a distance that is larger than a dimension value of an outer diameter of the electric wires; and

a width of the routing region surface in the first direction is set larger than the dimension value of the outer diameter of the electric wires.

2. The electronic control unit according to claim **1**, wherein:

the terminal blocks have a wire passage groove allowing a second electric wire to pass, which is different from the electric wire connected to the terminals supported by the terminal blocks; and

the routing region surface is a bottom surface of the wire passage groove.

3. The electronic control unit according to claim **1**, comprising connectors to which a second end portion opposite to the first end portion of each of the electric wires is connected.

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